

Vector Fields and Line Integrals

Lecture 03

Instructor: Asgher Ali

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Path Independence

Is the path along which we integrate is important?

- A GRAVITATIONAL FIELD $\mathbf{G}$ is a vector field that represents the effect of gravity at a point in space due to the presence of a massive object. The gravitational force on a body of mass m placed in the field is given by $\mathbf{F} = m\mathbf{G}$.
- An ELECTRIC FIELD $\mathbf{E}$ is a vector field in space that represents the effect of electric forces on a charged particle placed within it. The force on a body of charge q placed in the field is given by $\mathbf{F} = q\mathbf{E}$.
- In gravitational and electric fields, the amount of work it takes to move a mass or charge from one point to another depends on the initial and final positions of the object, not on which path is taken between these positions.

Definition

Let $\mathbf{F}$ be a vector field defined on an open region $\mathcal{D}$ in space, and suppose that for any two points A and B in $\mathcal{D}$, the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ along a path C from A to B in $\mathcal{D}$ is the same over all paths from A to B . Then the integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is **path independent in $\mathcal{D}$** and the field is conservative on $\mathcal{D}$.

Definition

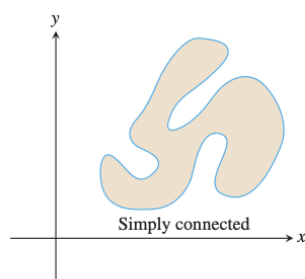
A vector field $\mathbf{F}$ is **conservative** if $\mathbf{F} = \vec{\nabla} f$ for some scalar function f .

Definition

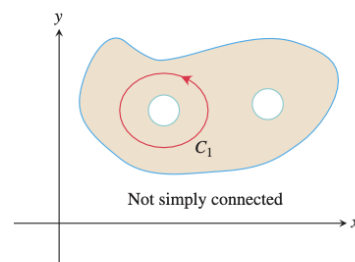
If $\mathbf{F}$ is a vector field defined on $\mathcal{D}$ and $\mathbf{F} = \vec{\nabla} f$ for some scalar function f on $\mathcal{D}$, then f is called a **potential function** for $\mathbf{F}$.

Assumptions and important geometrical notions

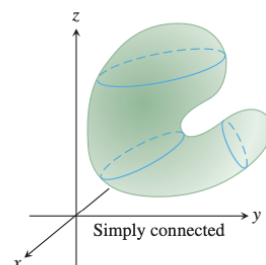
1. The curves we consider are **piecewise smooth**. Such curves are made up of finitely many smooth pieces connected end to end.
2. We consider vector fields $\mathbf{F}$ whose components have continuous first partial derivatives.
3. The domains $\mathcal{D}$ are **connected**.
4. The region $\mathcal{D}$ is **open** if two points in $\mathcal{D}$ can be joined by a smooth curve that lies in the region.
5. Some results needs $\mathcal{D}$ to be **simply connected**, which means that every loop in $\mathcal{D}$ can be contracted to a point in $\mathcal{D}$ without ever leaving $\mathcal{D}$.
6. Think of **connected** region as being "one piece" and **simply connected** region as not having any "loop catching holes".



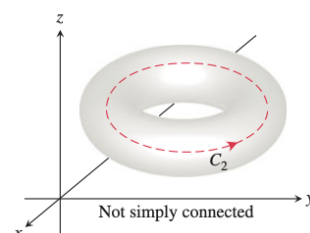
(a)



(c)



(b)



(d)

Fundamental theorem of line integral

Assume that $\mathbf{F} = \vec{\nabla} f$ on a domain $\mathcal{D}$.

- If $\mathbf{r}(t)$ is a path along a curve C from P to Q , then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(Q) - f(P)$$

- The **circulation** around a closed curve $P = Q$ is zero:

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0$$

Proof: Let $\mathbf{r}(t)$ be a path along the curve C in $\mathcal{D}$ for $a \leq t \leq b$ with $\mathbf{r}(a) = P$ and $\mathbf{r}(b) = Q$. Then

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \vec{\nabla} f \cdot d\mathbf{r} \\ &= \int_a^b \vec{\nabla} f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \end{aligned}$$

Chain Rule for Paths

If f and $\mathbf{r}(t)$ are differentiable function, then

$$\frac{d}{dt} f(\mathbf{r}(t)) = \vec{\nabla} f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$$

Explicitly, in case of three variables, if $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, then

$$\begin{aligned} \frac{d}{dt} f(\mathbf{r}(t)) &= \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \cdot \langle x'(t), y'(t), z'(t) \rangle \\ &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \end{aligned}$$

This implies that

$$\frac{d}{dt}f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$$

By applying Fundamental Theorem of Calculus

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_a^b \frac{d}{dt}f(\mathbf{r}(t)) \\ &= f(\mathbf{r}(t)) \Big|_a^b \\ &= f(\mathbf{r}(b)) - f(\mathbf{r}(a)) \\ &= f(Q) - f(P)\end{aligned}$$

It also proves path independence, because the quantity $f(Q) - f(P)$ depends on the end points but not on the path $\mathbf{r}$. If $\mathbf{r}$ is a closed path, then $P = Q$ and $f(Q) - f(P) = 0$

Example

Let $\mathbf{F}(x, y, z) = \langle 2xy + z, x^2, x \rangle$.

- Verify that $f(x, y, z) = x^2y + xz$ is a potential function.
- Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is a curve from $P = (1, -1, 2)$ to $Q = (2, 2, 3)$.

Theorem

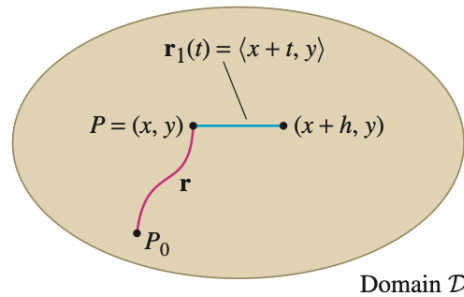
A vector field $\mathbf{F}$ on an open connected domain $\mathcal{D}$ is path independent if and only if it is conservative.

Proof: We have already proved that the conservative vector fields are path independent (Fundamental theorem of line integral).

Now, we assume that $\mathbf{F}$ is path-independent and prove that F has a potential function (F is conservative i.e., $\mathbf{F} = \nabla f$).

To simplify the notation, we treat the case of a planar vector field $\mathbf{F} = \langle F_1, F_2 \rangle$. The proof for vector fields in $\mathbb{R}^3$ is similar. Choose a point P_0 in $\mathcal{D}$, and for any point $P = (x, y) \in \mathcal{D}$, define

$$f(P) = f(x, y) = \int_C \mathbf{F} \cdot d\mathbf{r}$$



where $\mathbf{r}$ is any path in $\mathcal{D}$ from P_0 to P . Note that this definition of $f(P)$ is meaningful only because we assume that the line integral does not depend on the path $\mathbf{r}$.

We will prove that $\mathbf{F} = \nabla f$, which involves showing that $\frac{\partial f}{\partial x} = F_1$ and $\frac{\partial f}{\partial y} = F_2$.

We will only verify the first equation, as the second can be checked in the similar manner.

Construction: Let $\mathbf{r}_1$ be the horizontal path $\mathbf{r}_1(t) = \langle x + t, y \rangle$, for $0 \leq t \leq h$. For small h , $\mathbf{r}_1$ lies inside $\mathcal{D}$.

Let $\mathbf{r} + \mathbf{r}_1$ denote the path $\mathbf{r}$ followed by $\mathbf{r}_1$. It begins at P_0 and ends at $(x + h, y)$, so

Mathematical treatment:

$$\begin{aligned} f(x + h, y) &= \int_{\mathbf{r} + \mathbf{r}_1} \mathbf{F} \cdot d\mathbf{r} \\ &= \int_{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r} + \int_{\mathbf{r}_1} \mathbf{F} \cdot d\mathbf{r} \end{aligned}$$

Now

$$\begin{aligned} f(x + h, y) - f(x, y) &= \left(\int_{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r} + \int_{\mathbf{r}_1} \mathbf{F} \cdot d\mathbf{r} \right) - \int_{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r} \\ &= \int_{\mathbf{r}_1} \mathbf{F} \cdot d\mathbf{r} \end{aligned}$$

The path $\mathbf{r}_1$ has a tangent vector $\mathbf{r}'_1(t) = \langle 1, 0 \rangle$, which implies

$$\begin{aligned} \mathbf{F}(\mathbf{r}_1(t)) \cdot \mathbf{r}'_1(t) &= \langle F_1(x + t, y), F_2(x + t, y) \rangle \cdot \langle 1, 0 \rangle \\ &= F_1(x + t, y) \end{aligned}$$

This implies that

$$f(x + h, y) - f(x, y) = \int_{\mathbf{r}_1} \mathbf{F} \cdot d\mathbf{r} = \int_0^h F_1(x + t, y) dt$$

Using substitution, $u = x + t$, which will transform the lower limit to x and upper limit to $x + h$

$$f(x + h, y) - f(x, y) = \int_x^{x+h} F_1(u, y) du$$

In the next step,

$$\frac{f(x + h, y) - f(x, y)}{h} = \frac{1}{h} \int_x^{x+h} F_1(u, y) du$$

The integral on the right hand side is an average value of $F_1(u, y)$ on the interval $[x, x + h]$.

Mean value theorem for integrals

If $f(x)$ is continuous over an interval $[a, b]$, then there is at least one point $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$$

By using theorem above

$$\frac{f(x+h, y) - f(x, y)}{h} = F_1(c, y)$$

Applying limit on both sides

$$\lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = \lim_{h \rightarrow 0} F_1(c, y)$$

In addition, since c is between x and $x+h$, as h approaches 0, c approaches x , this implies

$$\begin{aligned} \lim_{h \rightarrow 0} F_1(c, y) &= \lim_{c \rightarrow x} F_1(c, y) \\ &= F_1(x, y) \end{aligned}$$

The final form is

$$\lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} = F_1(x, y)$$

$$\frac{\partial f}{\partial x} = F_1(x, y)$$

Inquiry Question

- How do we know whether a given vector field is conservative?
- If $\mathbf{F}$ is infact conservative, how do we find a potential function f such that $\mathbf{F} = \vec{\nabla} f$?

Component Test for Conservative Fields

Let

$$\mathbf{F} = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$

be a field on an open simply connected domain whose component functions have first partial derivatives. Then $\mathbf{F}$ is **conservative** if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z} \quad (1)$$

$$\frac{\partial P}{\partial x} = \frac{\partial M}{\partial z} \quad (2)$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad (3)$$

Example

Show that $\mathbf{F} = (e^x \cos y + yz)\mathbf{i} + (xz - e^x \sin y)\mathbf{j} + (xy + z)\mathbf{k}$ is conservative over its natural domain and find a potential function for it.